Mathematical Structures of QM

Homework(5)

B98xxx018 物理三 [illegible]

DATE

1D

1. p-rep

$$V(p'-p)=\frac{1}{2\pi\hbar}\int\left[-g\delta(x-a)\right]e^{i\frac{(p'-p)x}{\hbar}}dx=-\frac{g}{2\pi\hbar}e^{i\frac{(p'-p)a}{\hbar}}$$

$$\therefore \frac{p^2}{2m}\psi(p)-\frac{g}{2\pi\hbar}e^{-i\frac{pa}{\hbar}}\int e^{i\frac{p'a}{\hbar}}\psi(p')dp'=E\psi(p)$$

$$\Rightarrow e^{i\frac{pa}{\hbar}}\psi(p)=\frac{\frac{g}{2\pi\hbar}}{\frac{p^2}{2m}-E}\int e^{i\frac{p'a}{\hbar}}\psi(p')dp'$$

$$\Rightarrow \int e^{i\frac{pa}{\hbar}}\psi(p)\overset{dp}{=}\frac{mg}{\pi\hbar}\int\frac{dp}{p^2+p_0^2}\int e^{i\frac{p'a}{\hbar}}\psi(p')dp',\quad p_0\equiv\sqrt{-2mE}$$

$$\Rightarrow 1=\frac{mg}{\pi\hbar}\frac{1}{p_0}\underbrace{\tan^{-1}x\Big|_{-\infty}^{\infty}}_{\pi}\Rightarrow \boxed{E=-\frac{mg^2}{2\hbar^2}}$$

Let $\psi(p)=A\frac{e^{i\frac{pa}{\hbar}}}{p^2+p_0^2}$, to normalize it:

$$\int_{-\infty}^{\infty}\psi^*(p)\psi(p)dp=1\Rightarrow |A|^2\int_{-\infty}^{\infty}\frac{dp}{(p^2+p_0^2)^2}=|A|^2\frac{1}{p_0^3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2\theta\,d\theta$$

$$=\frac{\pi}{2p_0^3}|A|^2$$

$$\therefore A=\sqrt{\frac{2p_0^3}{\pi}}$$

$$\therefore \boxed{\psi(p)=\sqrt{\frac{2p_0^3}{\pi}}\frac{e^{i\frac{pa}{\hbar}}}{p^2+p_0^2}},\quad p_0=\frac{mg}{\hbar}$$

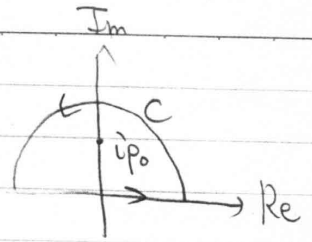
x-rep

$$\psi(x)=\underbrace{\frac{1}{\sqrt{2\pi\hbar}}}\int_{-\infty}^{\infty}\psi(p)e^{i\frac{px}{\hbar}}dp=\frac{1}{\sqrt{2\pi\hbar}}\sqrt{\frac{2p_0^3}{\pi}}\int_{-\infty}^{\infty}\underbrace{\frac{e^{i\frac{p(x-a)}{\hbar}}}{p^2+p_0^2}}_{\|}dp$$

$f(p)$

PAGE
DATE

① Consider $x-a>0$



$$|f(p)p| = \left| \frac{e^{i\frac{(x-a)}{\hbar}R(\cos\theta+i\sin\theta)}}{p^2+p_0^2} p \right|$$

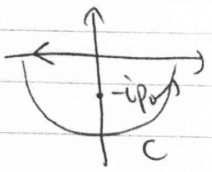
$$= \frac{Re^{-\frac{(x-a)R}{\hbar}\sin\theta}}{|p^2+p_0^2|}$$

for loop C, $\sin\theta>0$ $\therefore |f(p)p| \to 0$ as $R\to\infty \Rightarrow \int_C f(p)dp \to 0$

$\therefore \oint f(p)dp = 2\pi i \operatorname{Res} f(ip_0)$

$$\Rightarrow \int_C^{\to 0} + \int_{-\infty}^{\infty} = 2\pi i \frac{e^{-\frac{(x-a)}{\hbar}p_0}}{ip_0+ip_0} \Rightarrow \int_{-\infty}^{\infty} f(p)dp = \frac{\pi}{p_0} e^{-\frac{(x-a)}{\hbar}p_0}$$

② Consider $x-a<0$



Can also get $\int_C \to 0$ as $R\to\infty$

$$\Rightarrow \int_{-\infty}^{\infty} f(p)dp = -2\pi i \operatorname{Res} f(-ip_0)$$

$$= -2\pi i \frac{e^{\frac{(x-a)}{\hbar}p_0}}{-ip_0-ip_0} = \frac{\pi}{p_0} e^{\frac{(x-a)}{\hbar}p_0}$$

$\therefore \int_{-\infty}^{\infty} f(p)dp = \frac{\pi}{p_0} e^{-\frac{|x-a|}{\hbar}p_0}, \ \forall x \in \mathbb{R}$

$\therefore \psi(x) = \frac{p_0}{\pi}\sqrt{\frac{p_0}{\hbar}} \cdot \frac{\pi}{p_0} e^{-\frac{|x-a|}{\hbar}p_0} \Rightarrow \boxed{\psi(x) = \sqrt{\frac{p_0}{\hbar}} e^{-\frac{|x-a|}{\hbar}p_0}}$

10

2. $\psi_m(x) = \left(\frac{1}{\sqrt{2\pi}\Delta x}\right)^{1/2} e^{\frac{(x-\langle x\rangle)^2}{4(\Delta x)^2} + i\frac{x\langle p\rangle}{\hbar}}$

$$\therefore \psi_m(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi_m(x)\, e^{-i\frac{px}{\hbar}}\, dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \frac{1}{\sqrt{\sqrt{2\pi}\Delta x}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{4(\Delta x)^2} + \left(\frac{\langle x\rangle}{2(\Delta x)^2} + i\frac{\langle p\rangle - p}{\hbar}\right)x - \frac{\langle x\rangle^2}{4(\Delta x)^2}} dx$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \frac{1}{\sqrt{\sqrt{2\pi}\Delta x}} \sqrt{\frac{\pi}{\frac{1}{4(\Delta x)^2}}}\, e^{\frac{\left[\frac{\langle x\rangle}{2(\Delta x)^2} + i\frac{\langle p\rangle - p}{\hbar}\right]^2}{4\cdot\frac{1}{4(\Delta x)^2}} - \frac{\langle x\rangle^2}{4(\Delta x)^2}}$$

$$= \sqrt{\frac{2\Delta x}{\sqrt{2\pi}\hbar}}\, e^{-\frac{(p-\langle p\rangle)^2}{\hbar^2/\Delta x^2} + 2i\frac{\langle p\rangle - p}{\hbar}\frac{\langle x\rangle}{2\Delta x^2}\Delta x^2 + \Delta x^2\frac{\langle x\rangle^2}{4(\Delta x)^4} - \frac{\langle x\rangle^2}{4(\Delta x)^2}}$$

$$= \left(\frac{1}{\sqrt{2\pi}\Delta p}\right)^{1/2} e^{-\frac{(p-\langle p\rangle)^2}{4\Delta p^2} - \frac{i}{\hbar}(p-\langle p\rangle)\langle x\rangle}$$

$\nearrow \Delta x \Delta p = \frac{\hbar}{2}$